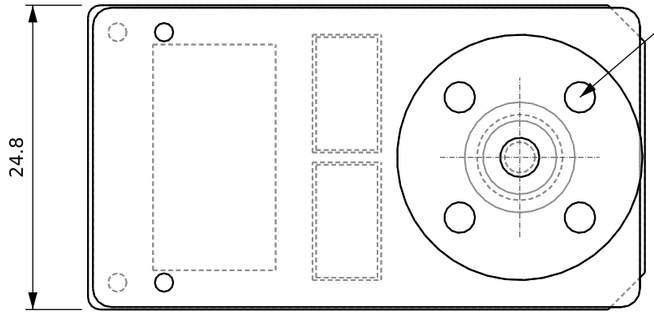
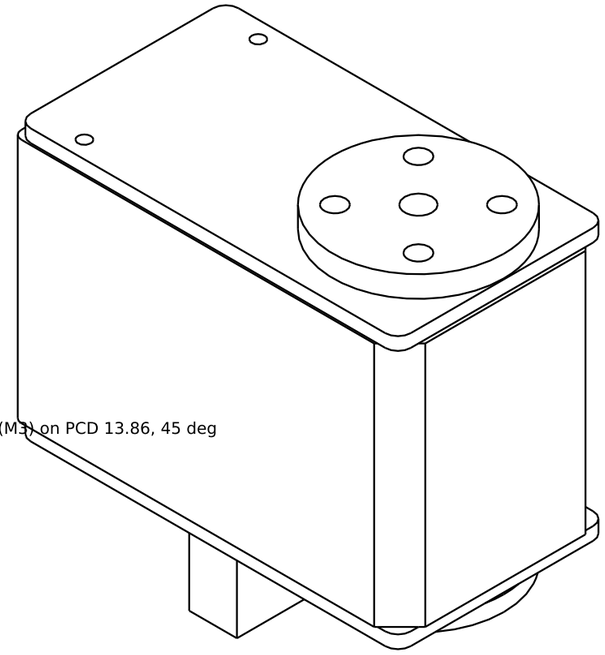
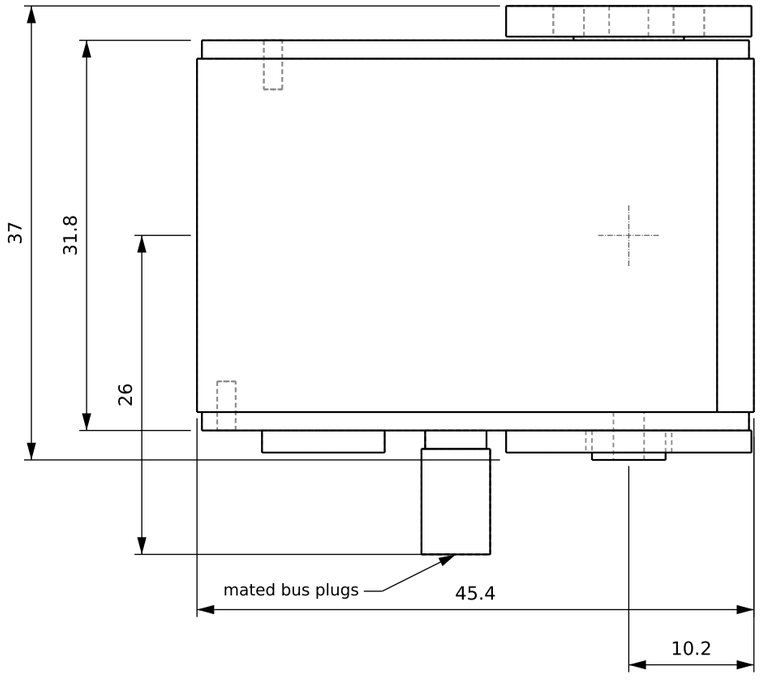


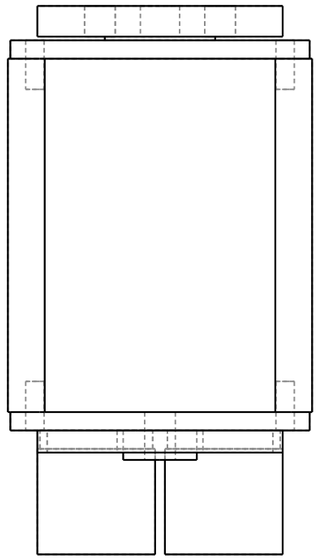
TOP



ISOMETRIC (NOT TO SCALE)



FRONT



RIGHT

FEATURES (servo frame)

1. Body: box x -35.2..10.2, y -12.4..12.4, z -14.4..14.4 (45.4 x 24.8 x 28.8). +X end vertical corners chamfered 3, -X end corners R1.
2. Caps: 44.6 x 24.4 x 1.5, corners R1.5, centred on x = -12.5, at z 14.4..15.9 and -15.9..-14.4.
3. Back pad (connector face): x -29.9..-19.9, y -9.2..9.2, z -17.7..-15.9.
4. Bus connector headers: x -16.6..-11.6, y 0.6..9.8 and -9.8..-0.6, z -17.4..-15.9. Mated plugs: x -16.9..-11.3, y 0.4..10.0 and -10.0..-0.4, z -26.0..-17.4.
5. Spline boss dia 9, z 14.7..16.2. Idler boss dia 6, z -18.3..-14.4, tapped dia 2.5 (M3).
6. Output horn dia 20, z 16.2..18.7: 4x dia 2.5 (M3) on PCD 13.86 at 45/135/225/315 deg, centre dia 3.2.
7. Passive horn dia 20, z -17.7..-15.9, centre dia 7.
8. Case holes dia 1.5 x 4 deep (M2 self-tapping): shaft face at (x, y) = (-29.0, +/-10.2); connector face at (-32.8, +/-10.2).

   SERVO POCKET P (cut this into a part, in the servo frame; 0.3 clearance included):

   a) box x -35.5..10.5, y -12.7..12.7, z -16.2..16.2

   b) back pad box x -30.2..-19.6, y -9.5..9.5, z -18.0..-15.5

   c) plug bay box x -17.6..-10.6, y -10.6..10.6, z -28.5..-15.5

   d) idler clearance cylinder dia 21.6, z -18.5..-15.0, on the shaft axis

   e) horn clearance cylinder dia 21.6, z 15.0..19.5, on the shaft axis

NOTES

- Measured from the SO-ARM100 STS3215 reference model and Feetech's drawing. Check the connector position and case-hole depth on a real servo.
- Each part sheet says where its pocket's servo frame sits and how it is rotated.

REFERENCE: in work sets A and B

THIRD ANGLE

PROJECT TALOS 5-axis arm	EST. MASS -
PART **STS3215 servo + pocket envelope (reference)**	SCALE 2:1
PART NO. **TAL-001**	UNITS / REV mm / A
QTY / ARM 6	DATE 2026-11-03
MATERIAL bought part	SHEET 2 of 12